

AI Circuit Breaker:

Section 1 - The Framework

This is a robust, metrology-driven architecture designed to underwrite trust in autonomous systems through deterministic validation. The framework consists of 5 separate subsystems, each of which requires lifecycles, design time construction, and run time maintenance. Below is a breakdown of components that need to be built out for each “subsystem”.

1. Subsystem - Semantic Knowledge Layer (Deterministic Knowledge)

This layer establishes the "Immutable Truth" that constrains probabilistic AI.

- **Vocabularies & Taxonomies:** Standardized terminology and hierarchical classifications for domain-specific entities (e.g., Telecom equipment types, signal states).
- **Domain Ontologies:** Formal representations of properties and relationships between entities, defining the physical and logical "laws" of the system.
- **Federated Graph Topology:** A distributed architecture allowing domain-specific graphs (e.g., Optical Transport, 5G Core) to operate independently while remaining interconnected.
- **Unifying Ontology:** A top-level "Global" ontology that maps across federated graphs to support multi-directional search and seamless navigation between disparate domains.
- **Logic Reasoners:** Inference engines that use **Agentic AI** to evaluate whether proposed actions align with the ontological rules.
- **Complementary Vector DBs:** High-speed retrieval systems that store semantic embeddings, allowing Agentic AI to quickly reference similar historical contexts.

2. Subsystem - Context Engine (Relevancy at Scale)

This layer refines the knowledge base into a real-time, actionable situational model.

- **Relevancy Mapping Engine:** Uses Agentic AI to filter the Knowledge Graph for nodes and relationships specifically relevant to the current operational window.
- **Outcome Optimizer:** Leverages **Persona and Profile Models** (e.g., "North Shore 5G Node Profile") to simulate the impact of actions on specific user segments.
- **Digital Twin Synchronizer:** (Additional Component) Ensures the internal context model is perfectly synced with real-time deterministic sensor data.

- **Environment Stress Monitor:** (Additional Component) Tracks external variables (weather, power fluctuations) that alter the situational risk profile.

3. Subsystem - AI Circuit Breaker (The Veto Gate)

The deterministic gatekeeper that compares AI intent against ground truth.

- **Semantic Guardian Logic:** The first line of defense; it "vetos" any intent that violates the fundamental rules in the Knowledge Subsystem.
- **Multi-Layer Validation Logic:** Performs cross-checks across sensors, logical states, and historical success patterns.
- **Decision Logic - Intent Mapping:** Maps the AI's "Fuzzy" intent into a precise vector for mathematical comparison
 - $(|Intent - Reality|)$.
- **Decision Rules & Breaker Thresholds:** Pre-defined engineering tolerances; if the **Delta** exceeds these thresholds, the circuit "trips."
- **Metrology for Ethics & Bias:** (Additional Component) Monitors for disparate impact or drift from human-defined ethical alignment.
- **Drift, Accuracy, & Hallucination Monitors:** Continuous tracking of **Mean Time Between Hallucinations (MTBH)** and system accuracy.

4. Subsystem - Recursive Learning & Model Governance

The "Immunology" layer that prevents the system from re-learning errors.

- **Data Collection & Sensor State:** Captures the full state of the network at the moment of execution.
- **Positive Learning:** Ingests successful, validated actions to reinforce the model.
- **Negative Learning (Exclusion Filter):** Specifically flags "Tripped" data and hallucinations to ensure they are **purged** from training sets, preventing **Model Collapse**.
- **Scenario-Specific Vector DB Updates:** Modifies domain-specific vector databases to include new edge-case embeddings discovered during operation.
- **Ontology Enhancement:** Automatically proposes new term nodes and relationships to the Knowledge Subsystem based on verified operational discoveries.

5. Subsystem - Human Interaction with AI Training & Learning

The interface for the "Human Trustee" to underwrite system reliability.

- **Credibility & Trust Assessment:** Visualizes the system's "Trust Score" based on historical MTBH and current Value-Drift.
- **Mean Time to Hallucination (MTTH/MTBH):** A primary metric showing the average operational time between high-confidence errors.
- **Cognitive Load Monitor:** (Additional Component) Tracks the human trustee's state (e.g., via HRV) to ensure they are capable of oversight.
- **Underwriting Console:** (Additional Component) Allows the trustee to formally "Sign Off" on or "Audit" new recursive learning updates.

Additional Considerations:

System Data Flows

Strategic Flow: Human Intent → Knowledge Subsystem → Strategic AI → **Proposed Action Vector**.

Metrology Flow: Real-World Sensors → Context Subsystem → **Ground Truth Vector**.

Validation Flow: Proposed Action Vector + Ground Truth Vector → **Circuit Breaker (Delta Analysis)**.

Governance Flow: Breaker Trip/Success → Data Exclusion Filter → **Recursive Learning Pipeline** → Knowledge Update.

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Section 2 - Trustworthiness Metrology & Semantic Governance

Below are a few of the key measures we'll need to define and establish in the framework. These measures provide the underwriting for trust within autonomous systems. The measures anchor probabilistic AI behavior to deterministic engineering constants through real-time telemetry and a "Veto-First" architecture (Breaker).

Semantic Anomaly Score (S_a)

Purpose: Quantifies the logical divergence between AI-proposed actions and the established domain ontology.

- **Inputs:** * Proposed Intent Vector ($\vec{I}$) from the Strategic AI.
 - Local Semantic Neighborhood (N_o) from the **Federated Graph Topology**.
- **Transform:**
$$S_a = 1 - \frac{\vec{I} \cdot \vec{N}_o}{\|\vec{I}\| \|\vec{N}_o\|}$$
- (Where $\vec{N}_o$ represents the centroid of the valid ontological state space).
- **Output:** Scalar value [0, 1].
- **Tolerance:** $S_a < 0.15$. Scores ≥ 0.15 trigger an immediate **Semantic Veto**.

2. Contextual Relevancy Index (C_r)

Purpose: Evaluates if the AI's situational model is grounded in real-time environmental data.

- **Inputs:** * Situational Context Model (C_m) from the **Context Engineering Subsystem**.
 - Sensor Ground Truth State (S_{gt}) from **Deterministic Sensors** (LiDAR/OTN probes).

- **Transform:**

$$C_r = \sum_{i=1}^n w_i (1 - |C_{m,i} - S_{gt,i}|)$$

(Where w_i is the importance weight of the i -th environmental variable).

- **Output:** Scalar value [0, 1].
- **Tolerance:** $C_r > 0.85$. Any value below this threshold forces the system into a **Conservative Safe State**.

Mean Time Between Hallucinations (MTBH)

Purpose: Primary reliability metric for longitudinal AI health tracking.

- **Inputs:** * Total Operational Time (T_{ops}).
 - Count of Verified Hallucinations/Circuit Breaker Trips (N_h).
- **Transform:**

$$MTBH = \frac{T_{ops}}{N_h}$$
- **Output:** Time duration (hours/days).
- **Tolerance:** Minimum MTBH varies by domain (e.g., 500 hours for 5G Core operations).

Human-AI Calibration Coefficient (K_{trust})

Purpose: Synchronizes system sensitivity with the operator's physiological state.

- **Inputs:** * Human Cognitive Load (L_h) derived from HRV/EEG data.
 - AI Health Indicator (H_{ai}) derived from S_a and C_r trends.
- **Transform:**

$$K_{trust} = \frac{H_{ai}}{L_h}$$
- **Output:** Sensitivity Multiplier for Breaker Thresholds.
- **Tolerance:** If $K_{trust} < 0.5$ (High human stress + Low AI health), the **Circuit Breaker** initiates an autonomous lockdown.

Prompt-to-Ontology Traceability Logic

Purpose: Verification of prompt engineering against deterministic "Knowledge Warehousing".

- **Logic Process:**
 1. **Extract Entity:** Parse the Prompt for operational nodes.
 2. **Verify Lineage:** Trace the node to the **Federated Graph Topology**.
 3. **Cross-Reference:** Compare prompt-defined constraints with the **Domain-Specific Vector DB** history.
- **Completeness Audit:** Ensures all decision weights are traced to a deterministic source of truth.

Subsystem Connectivity & Calibration

The **Real-Time Calibration** engine ensures that all subsystems are synchronized at a frequency of 100Hz.

- **Subsystem Sync:** Knowledge Base updates are broadcast to the **Context Modeler** to ensure the "Relevancy Mapping" remains current.
- **Audit Trail:** Every autonomous decision is appended with a **Decision Traceability** hash, linking it to the specific sensor readings and ontological nodes used for verification.

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Section 3 - Metrics Definitions and Sources of Record

Below is a list of definitions and sources for metrics that feed into the above set of measures to support a rigorous engineering foundation for the **Trustworthiness Framework**, each metric is elaborated below with its mathematical grounding and specific data requirements. The examples below are related to Network Operations (source of familiarity in my case). We would need to generalize these and tailor them for each operational domain.

1. Semantic Anomaly Subsystem

This layer detects "Logical Drift"—when an AI's plan violates the fundamental physics or rules of the network.

- **Intent Vector ($\vec{I}$):**
 - **Definition:** A high-dimensional numerical representation (embedding) of the AI's proposed operational command. It captures the semantic "meaning" of the action (e.g., "Reroute 50Gbps traffic via Node-X").
 - **Data Sources:** Strategic AI output logs, **Domain-Specific Vector DB** embeddings, and gRPC-based intent manifests.
- **Local Semantic Neighborhood (N_o):**
 - **Definition:** The cluster of valid nodes, relationships, and "known-good" states within the **Federated Graph Topology** that surround the current operation. It represents the "reachable" and safe logical space.
 - **Data Sources:** **Federated Graph Topology** nodes, historical "success" logs, and real-time topology discovered via LLDP.
- **Centroid of the Valid Ontological State Space:**
 - **Definition:** The mathematical "Center of Gravity" of all safe engineering configurations. If the Intent Vector ($\vec{I}$) drifts too far from this center, it is statistically flagged as a hallucination.
 - **Data Sources:** Verified **Unifying Ontology** datasets and long-term historical network performance baselines.

2. Contextual Relevancy Subsystem

This layer ensures the AI isn't making "blind" decisions that ignore the current physical environment.

- **Situational Context Model (C_m):**
 - **Definition:** The AI's internal "mental map" of the world. It includes what the AI *thinks* is happening (e.g., assuming a link is available because it's not "software-down").
 - **Data Sources:** Virtualized Network Function (VNF) state tables, temporary configuration files, and recent prompt engineering context.
- **Sensor Ground Truth State (S_{gt}):**
 - **Definition:** The absolute physical reality as measured by deterministic hardware. It serves as the "Solid Benchmark" for all AV and autonomous models.
 - **Data Sources: Deterministic Sensors** (LiDAR, Optical Power Meters, RADAR, Fixed GNSS receivers).
- *i* – **th Environmental Variable:**
 - **Definition:** Specific external factors that influence the risk of an action, such as local weather patterns or physical fiber strain.
 - **Data Sources:** Weather station APIs (RADAR/SNR sensors), physical site security sensors, and localized hardware telemetry.

3. Longitudinal Reliability (MTBH)

- **Total Operational Time (T_{ops}):**
 - **Definition:** The total cumulative "Up-Time" of the autonomous agent.
 - **Data Sources:** System uptime logs, AWS/Google Cloud instance duration logs, and network controller timestamps.

4. Human-AI Calibration Subsystem

This layer synchronizes system safety with the status of the human operator.

- **Human Cognitive Load (L_h):**
 - **Definition:** The amount of mental effort and working memory resources invested in the task by the operator.
 - **Data Sources: HRV/EEG Data** (oscillations in heart rate intervals and frontal theta brain waves). This is probably a bit of a stretch given privacy concerns and the need to wear monitors by staff. We may opt for a simpler set of metrics based on duration at station, frequency of breaks, and other indicators for early onset of fatigue.
- **AI Health Indicator (H_{ai}):**
 - **Definition:** A fused score tracking the stability of the AI. It is primarily driven by S_a **Trends** (logic stability) and C_r **Trends** (environmental grounding).
 - **Data Sources:** Aggregated circuit breaker trip logs, drift avoidance values, and real-time anomaly detection scores.